Final Project Proposal

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Project Description

We chose to implement the Audio Sampler Project. We will design an App that allows users to edit and process audio signals within the MATLAB APP Designer interface. This application will work as a simple digital audio player that enables a complete workflow from loading and editing to synthesis and exporting.

The goal of this project is to build a GUI design, audio synthesis, signal processing, and event-driven programming. The sampler can arrange up to 9 audio samples into a 3 × 3 grid of buttons, and users can select a segment to edit and save the synthesized tone to play. It will also help manage audio files and visualize the signal.

General Description of Implementation

We will build the audio sampler by using MATLAB App Designer. The core interface features a 3*3 simple grids where users can load audio files via a file selector. The system will automatically display waveforms on the main axis.

The sample editing panel will involve volume control, reverse playback, speed adjustment, delay echo and a six-band equalizer. On the chopping board, we will provide Start and End time sliders for audio trimming. All modifications update the waveform display in real-time and can directly overwrite the original sample.

The synthesizer panel will support four waveform types, including sine, square, sawtooth, and AM-modulated waves. It will synthesize pure tones in real-time using mathematical functions. A generated signal will dynamically display its waveform on an axis. Users will be able to save it to the sample slot. All processed audio will support exporting. This Audio Sampler will

Reach Goals

1. Six-band Equalizer

An equalizer with six adjustable frequency bands (63 Hz, 125 Hz, 250 Hz, 500 Hz, 1 kHz, 2 kHz) will be included. Each slider will either increase or decrease the loudness of a particular frequency range which will permit users to control the audio's tonal balance.

2. Audio Export

Users can save their modified samples or synthesized tones as external audio files. This permits the application to serve as a basic instrument for the creation and exporting of edited sounds.

Responsibility and Timeline

Jiawei Zhu:

- ☑ ~~Editing proposal based on requirements~~
- ☐ Implement loading audio files into the 3x3 button grid.
- ☐ Implement playback logic for each loaded sample

- ☐ Implement the axis that displays the waveform of the currently playing audio sample with respect to time.
- ☐ Checking the loaded sample, synthesized tones, and effects, all work goes well without errors.

Ling Ai:

- ☑ ~~Editing proposal based on requirements~~
- ☑ ~~Implement tone synthesis (sine, square, sawtooth, AM) and save generated tones to sample slots~~
- ☑ ~~Plot synthesized waveforms on the second UIAxes and ensure correct updates~~

Xinyi Li

- ☑ ~~Build the GUI layout in App Designer~~
- ☑ ~~Editing proposal based on requirements~~
- ☐ Effect and sample modification panel: volume control, reverse playback, speed modification, echo/delay effect implementation
- ☐ Chopping panel: enable selecting samples and editing by user-defined start and end time; update the sample after chopping

Milestones

Task	Target Date
Proposal, General GUI	November 18
Core component done by each member, Group discussion	November 26
Putting components together, debugging	December 3 (Mid Project Check)
Polishing, testing, and further debugging	December 5
Final test, YouTube Demo and reflection	December 7

Projected Project Timeline

- 2 days for proposal and GUI layout
- 10 days for core audio loading functionality, by each member, individually
- 5~7 days for debugging, polishing, and testing
- 2~3 days for reflection and YouTube Demo
- Total expected time: 19–22 days (Debug and reach features perhaps need more time. It depends on what process we have. So it will be flexible.)
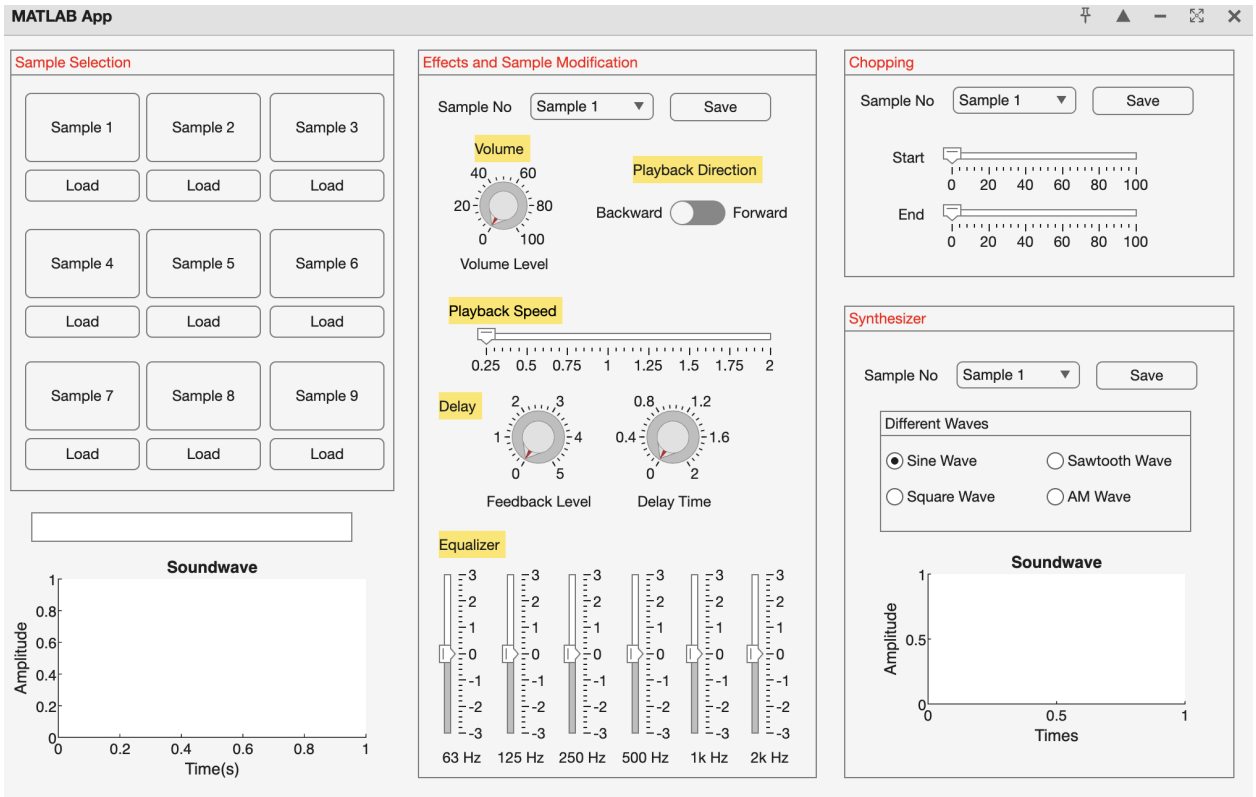
Image of Game Design